

BONE AND JOINT PATHOLOGY

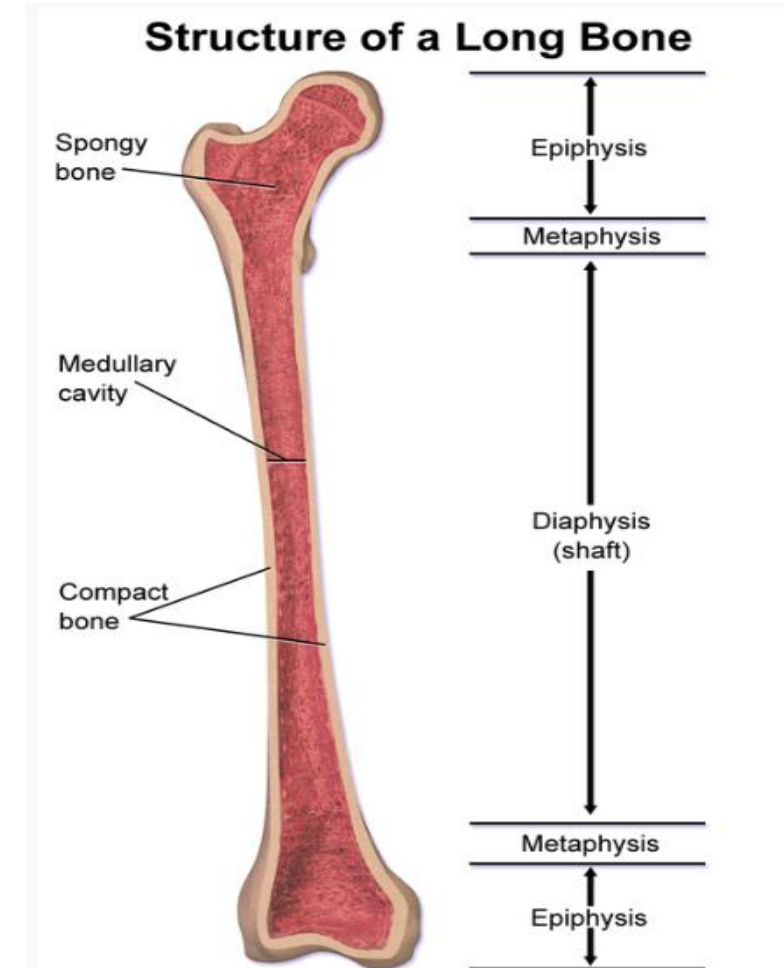
BY MELESE A. (MD, PATHOLOGIST)

Lecture outline

1. Congenital diseases of bone
2. Acquired diseases of bone development
 - a. Fractures
 - b. Osteonecrosis
 - c. Osteomyelitis
3. Metabolic bone disease
4. Tumors
5. Joint pathology

Introduction

- Musculoskeletal system is comprised of the bones, joints and the muscles.
 - Bones have three roles which are
 - Mechanical support
 - Mineral storage
 - Hematopoiesis
- Mechanical functions of bone include
 - Protection for the brain, spinal cord and chest organs
 - Rigid internal support for limbs
 - Deployment as lever arms in the skeletal muscle



Congenital Diseases of Bone

- Congenital diseases of bone range from **localized malformations** to **hereditary disorders** associated with abnormalities affecting the entire skeletal system.
- Can result in the **absence** of bones, **extra** bones, or **inappropriately fused** bones
- These are typically due to **mutations** in genes affecting localized migration and condensation of primitive mesenchymal cells.

1. Achondroplasia

- ❖ A syndrome of **short-limbed dwarfism** and **macrocephaly**
- ❖ Represents a **failure of normal epiphyseal cartilage formation**.
- ❖ It is the most common genetic form of dwarfism (1:15,000 live births)
- ❖ Is inherited as **autosomal dominant** trait.
- ❖ These people have **normal** mentation and life spans.

Pathogenesis

- ❖ Caused by mutations in the **fibroblast growth factor receptor 3 (FGFR3)** on **chromosome 16**.
 - inhibits **proliferation** and **differentiation of chondrocyte** which impedes growth plate development

1. Achondroplasia

Clinical features

- ❖ Have short stature
- ❖ The average height of an adult male is **131 cm** and adult female is **124cm**.
- ❖ Have an average-size trunk, **short arms and legs**
- ❖ Short arms have limited range of motion at the elbows
- ❖ A big head (macrocephaly) with prominent forehead
- ❖ Scoliosis and Kyphosis



2. Osteogenesis Imperfecta

- ❖ A group of genetic (mainly autosomal dominant) disorders of connective tissue due to mutations in the gene for **type I collagen**.
- ❖ It affects the **skeleton, joints, ears, ligaments, teeth, sclerae** and **skin**
- ❖ It is characterized by **hard and easily broken** bones often from little or no apparent cause.

Pathogenesis

- ❖ The pathogenesis involves mutations of **COL1A1** and **COL1A2** genes
 - Encode the **alpha1** and **alpha2** chains of **type I procollagen**, the major structural protein of bone.
 - These genes are located in chromosomes 17 and 7 respectively.
- ❖ Mutations cause substitution of other amino-acids for the glycine type I collagen at every third residue.

2. Osteogenesis Imperfecta

Clinical features

❖ This depends on the type of OI

Type 1

- ❖ Mildest form
- ❖ Multiple fractures when child begins walking and sitting
- ❖ Blue sclera (**thin sclera** with underlying choroids)
- ❖ Hearing abnormalities (fusion of ossicles) beginning in early 20s or 30s.
- ❖ Extremely thin and curved bones and bluish yellow teeth.



3. Osteogenesis Imperfecta

Type II

- ❖ Most severe form
- ❖ Numerous fractures and severe bone deformity
- ❖ Small stature with underdeveloped lungs.



2. Osteogenesis Imperfecta

Type III

- ❖ Short stature.
- ❖ Barrel-shaped rib cage.
- ❖ Bone deformity, often severe
- ❖ Brittle teeth and hearing loss

Type IV

- ❖ Similar to type I
- ❖ White sclera
- ❖ Bone cortex may mature during adolescence or afterwards



BONE PATHOLOGY

1. Congenital diseases of bone
2. Acquired diseases of bone development
 - a. Fractures
 - b. Osteonecrosis
 - c. Osteomyelitis
3. Metabolic bone disease
4. Tumors

Fractures

- ❖ The most common bone lesion
- ❖ Defined as a **discontinuity** of the bone

Pathogenesis

- ❖ The lesion typically results from blunt trauma to the muscle and soft tissues, usually of the lower limb; however, some occur **spontaneously**.
- ❖ Fractures can be classified as **traumatic**, **pathological** or **stress** fractures

Common causes

Trauma.

- A fall, a motor vehicle accident, or trauma during a football game can all result in fractures

Pathological fractures

- **Through abnormal bone:**
 - **Neoplastic:** Metastases (breast, lung, RCC, thyroid), myeloma, primary bone tumors
 - **Metabolic:** Osteoporosis, osteomalacia, Paget's disease
 - **Infectious:** Osteomyelitis
 - **Genetic:** Osteogenesis imperfecta, osteopetrosis

A stress fractures

- ❖ Refers to accumulation of stress-induced micro-fractures ,which eventually result in a true fracture throughout the bone cortex.
- ❖ They can result from repeated mechanical injury.

Fractures

Common types of fractures

❖ Stable fracture.

- The broken ends of the bone line up and are barely out of place.

❖ Open or compound fracture.

- The skin is pierced by the bone that breaks the skin at the time of the fracture.
- The bone may or may not be visible in the wound.



Fractures

Transverse fracture

- has a **horizontal** fracture line

Oblique fracture

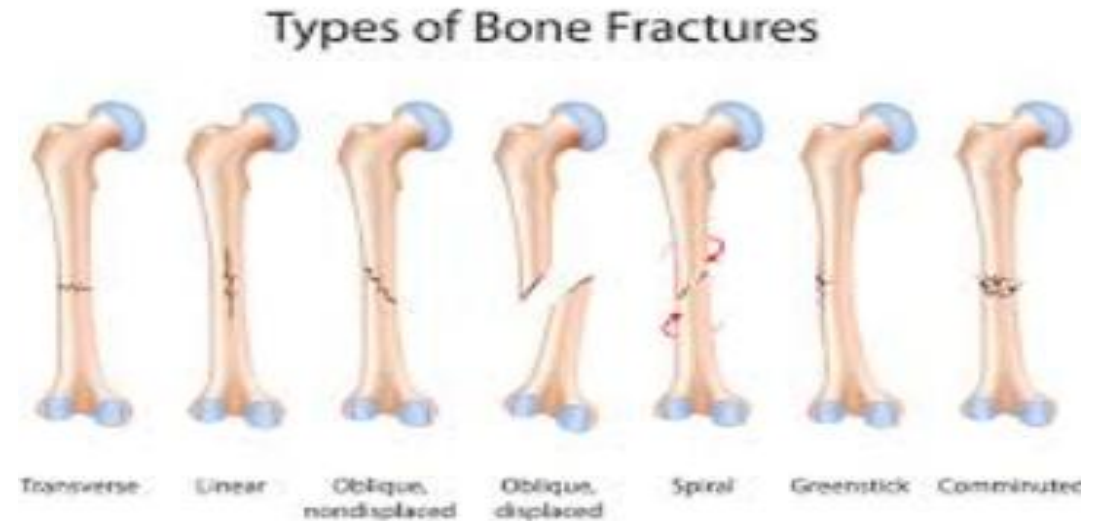
- has an **angled** pattern

Comminuted fracture

- the bone shatters into **three or more** pieces

Signs and Symptoms

- Pain on the affected area
- Unable to move the injured limb.
- Other common symptoms include:
 - Swelling and tenderness around the injury area
 - Bruising
 - Deformity - a limb may look "out of place" .



Osteonecrosis

- Refers to **death of bone** in the absence of infection.
- Causes/ Risk factors
 - Long-term steroid treatment
 - Alcohol abuse
 - Joint injuries
 - Having diseases such as arthritis and cancer



Osteonecrosis

Symptoms and Signs

- Patients may remain asymptomatic for weeks to months after the vascular insult.
- Pain then develops gradually with progressive collapse of the joint
- Pain is worsened by motion and weight bearing and relieved by rest

Osteomyelitis

- Osteomyelitis is an inflammation of the **bone and the bone marrow.**

Causes

- Commonly caused by **Staphylococcus aureus(Pyogenic Osteomyelitis)** and Mycobacterium tuberculosis (Tuberculous Osteomyelitis)
- Micro - organisms can enter a bone in 3 ways
 - Via the blood stream
 - From a nearby infection
 - Direct contamination
- **Children:** 80% of cases, peak 2-12 years
- **Adults:** Vertebrae most common site (lumbar > thoracic)

Osteomyelitis

Pathogenesis

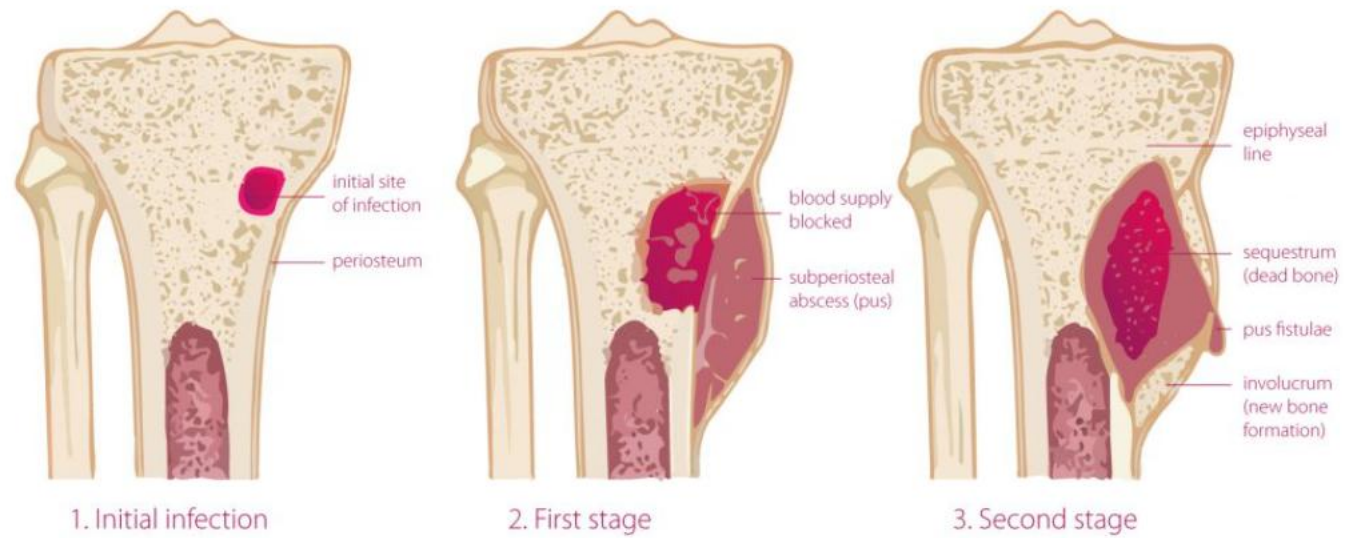
- ❖ Primarily affects the **metaphyseal area** because of **vascular supply** in this region
 - **Metaphysis** of long bones (distal femur, proximal tibia, proximal humerus)

Pathologic Sequence

- **Bacterial seeding** → acute inflammation
- **Increased intraosseous pressure** (rigid compartment)
- **Compression of vessels** → ischemia
- **Necrosis** → **sequestrum** formation (devitalized bone)
- **Periosteal elevation** → new bone formation (**involucrum**)
- **Draining sinus** through cortex (**cloaca**)

Signs and symptoms

- Fever or chills
- Irritability or lethargy in young children
- Localized bone pain
- Swelling, warmth and redness over the area of the infection
- Sometimes it causes no signs and symptoms or has signs and symptoms that are difficult to distinguish from other problems



development of osteomyelitis



Osteomyelitis

Complications

- Osteonecrosis - Bone death
- Septic arthritis
- Impaired growth
- Skin cancer

Tuberculosis of Bone

- Tuberculosis of bone originates from the lung or lymph-nodes, and reaches the bone by **hematogenous** spread.
- Often affects the spine, termed as **Tuberculous spondylitis** or **Pott disease**
 - Is a complication of childhood tuberculosis and affects the **bodies** of the vertebrae, with sparing of the lamina
- The **thoracic vertebrae** are most commonly affected

Osteomyelitis

Pathology

- The pathology in Tuberculous spondylitis is similar to tuberculosis at other sites.
- The granulomas first produce caseous necrosis of the bone marrow, which leads to slow resorption of bony trabeculae and, occasionally, to cystic spaces within the bone.
- Because there is little or no reactive bone formation, affected vertebrae usually collapse, leading to **kyphosis** and **scoliosis**.
- The intervertebral disk is crushed and destroyed by the compression fracture, rather than by invasion of micro-organism.



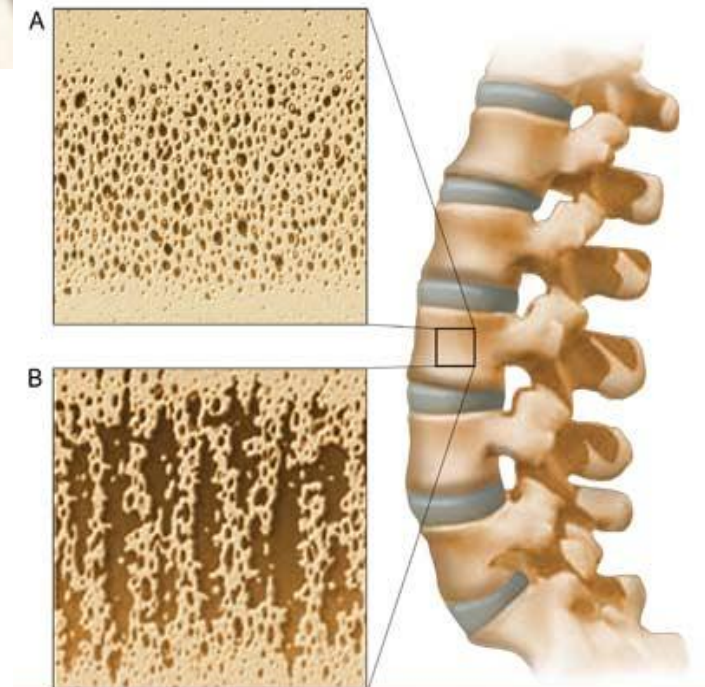
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Osteoporosis

- AKA “porous bone” is a disease characterized by **increased sponginess** of the bone resulting from **reduced bone mass**.

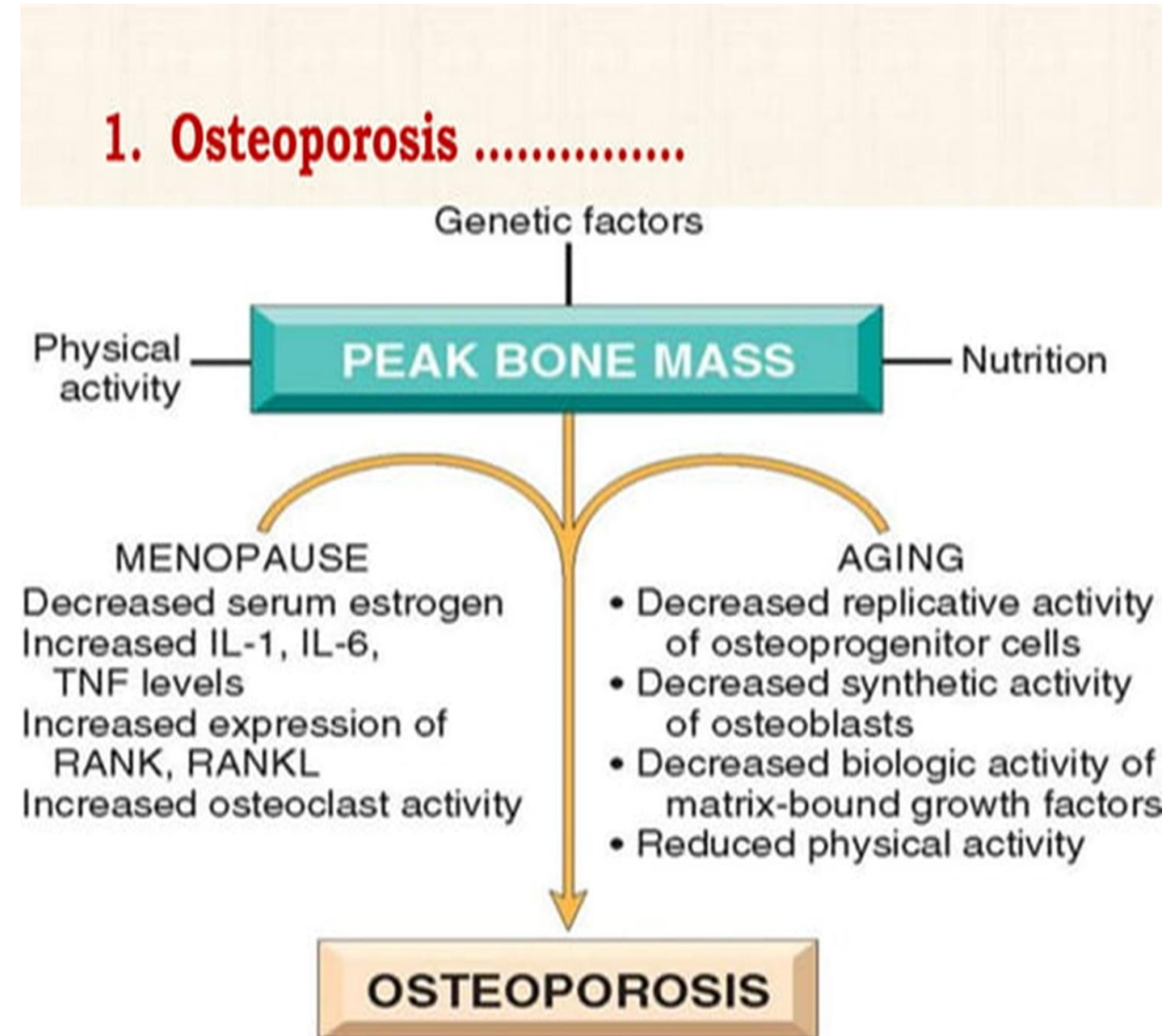
Classification

- Osteoporosis can be **primary** or **secondary** due to some other factor.
- Primary disorder can further be classified as
 - **Type 1 or postmenopausal osteoporosis**
 - Is associated with decreased levels of estrogen
 - Has a greater effect on **trabecular** than cortical bone
 - **Type 2 or senile osteoporosis**
 - Is a consequence of aging
 - Often augmented by inadequate calcium or Vit-D intake



Pathogenesis

- Osteoporosis occurs when bone loss exceeds bone formation, resulting in a low bone mass.
- **Failure of the remodeling cycle** to replace all the resorbed bone.



Risk Factors

- Sex (F > M)
- Physical inactivity
- Slender body build
- Race (Caucasians > African Americans)
- Early menopause
- Increasing age
- Calcium nutritional state: Deficient dietary intake

Signs and Symptoms

- Patients are asymptomatic unless a fracture has occurred.

Morphology

- The trabecular plates become perforated, thinned, and lose their interconnections, leading to progressive micro-fractures.

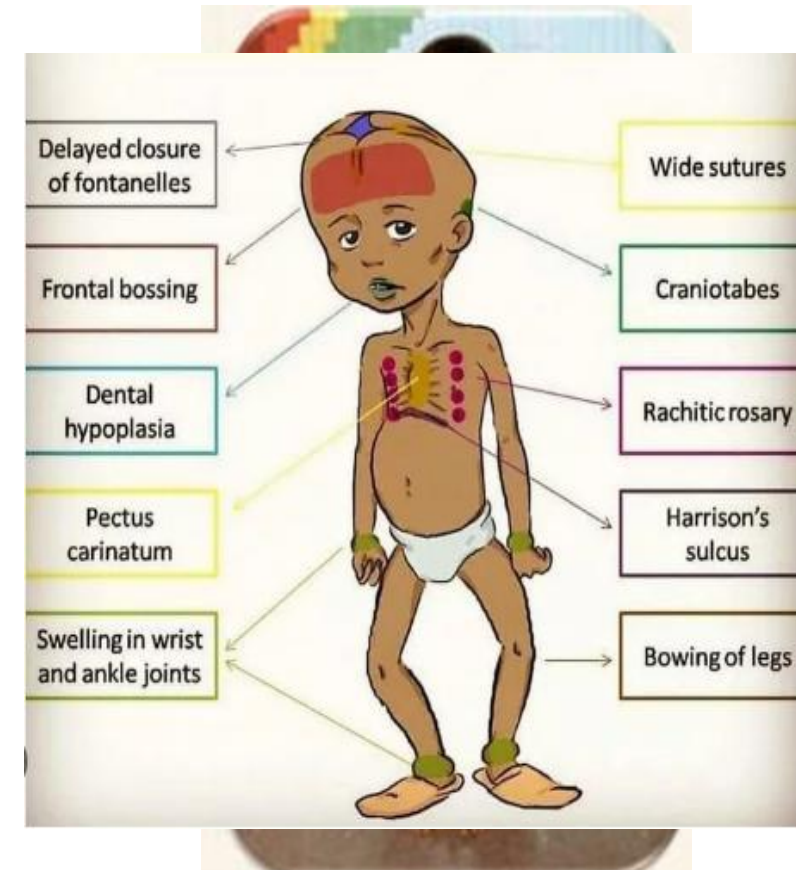


Osteomalacia and Rickets

- **Defect:** Impaired **mineralization** of osteoid (bone) and growth plate cartilage (rickets).
- **Osteomalacia**
 - a softening of bones, often caused by vitamin D deficiency in **adults**.
- **Rickets**
 - the softening and weakening of bones in **children**,
 - usually because of an extreme and prolonged vitamin D deficiency.

Pathogenesis

- Vitamin D promotes the absorption of calcium and phosphorus from the gastrointestinal tract.
- Deficiency of vitamin D makes it difficult to maintain proper calcium and phosphorus levels in bones, which can cause soft bones.
- Soft bones are more likely to **bow and fracture** than are harder and healthy bones.
- Sign and symptoms
 - The most common signs and symptoms are bowed legs



Paget disease (osteitis deformans)

- Is a **localized** disorder of **bone remodeling**, resulting in **resorption** followed by **excessive disorganized bone replacement**.
- Produces **thickened but weak** bone that is susceptible to deformity and fracture
- It is related to a **virus** infection with **paramyxovirus**, and also possibly to have genetic predisposition.
- begins after age 40 years and is common in those of European ancestry.

Pathogenesis

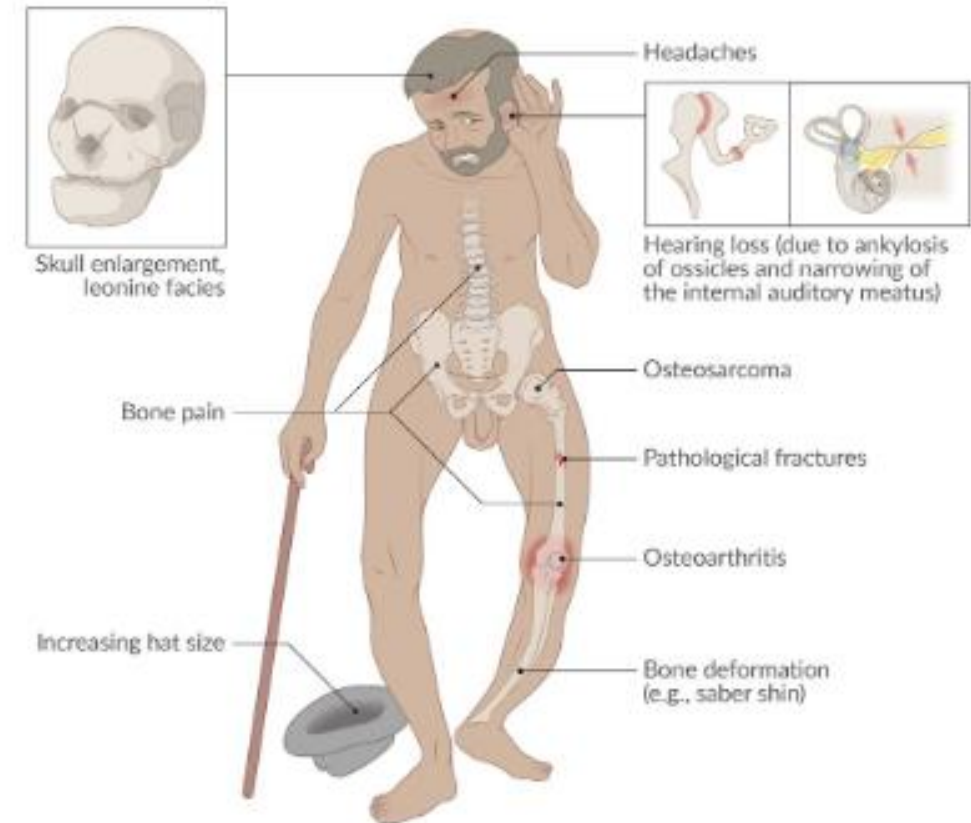
- Paget disease develops in 3 stages:
 - The osteolytic stage (osteoclastic activity predominates)
 - The mixed osteolytic-osteoblastic stage
 - The osteosclerotic stage (osteoblastic activity predominates in this burnout stage)

Pathology

- Involved bones are **thick but weak** and **fracture easily**.
- Skull involvement leads to increased head size and foraminal narrowing that can impinge on cranial nerves, often leading to **deafness**.
- Involvement of facial bones may produce a **lion-like face**.

Complications

- Arteriovenous shunts within marrow, which may result in **high-output cardiac failure**
- Increased incidence of **osteosarcoma** and other sarcomas.



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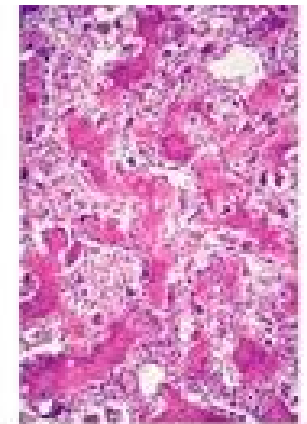
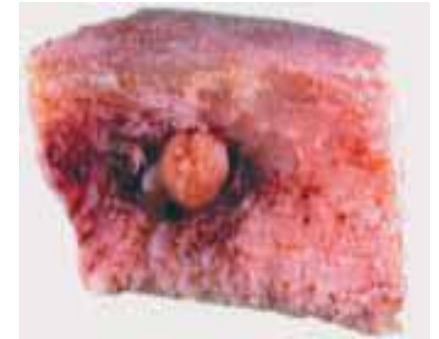
Benign tumors of bone

Osteoma

- Is a benign neoplasm that frequently involves the skull and facial bones.
- Osteoma can be associated with **Gardner syndrome**

Osteoid Osteoma

- is a benign, painful growth of the diaphysis of a long bone, often the tibia or femur.
- It affects males more than females, with peak age 5 to 25 years.
- The pain tends to be **worse at night** and **relieved by Aspirin**
- X-rays studies show **central radiolucency surrounded by a sclerotic rim**
- Microscopically, these tumors show
 - a small (<2cm) lesion of the cortex characterized by a central nidus of osteoid surrounded by dense sclerotic rim of reactive cortical bone.



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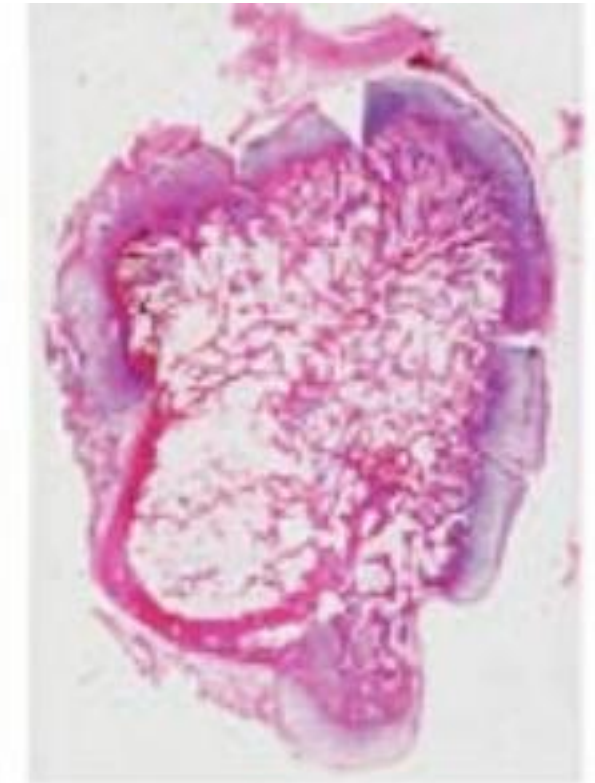
Benign tumors of bone

Osteoblastoma

- Osteoblastoma is a tumor that is similar to an osteoid osteoma but is larger (>2 cm) and often involves **vertebrae**.

Osteochondroma (exostosis)

- is a benign bony metaphyseal growth capped with cartilage
- originates from epiphyseal growth plate.
- presents in **adolescent males** who have firm, solita growths at the ends of long bones.
- Osteochondromas may be
 - Asymptomatic
 - Cause pain
 - Produce deformity or
 - Undergo transformation (rare)malignant



Cartilage

Bone

Marrow

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Osteosarcoma

- Is the most common primary malignant tumor of bone
- occurs more frequently in males than females.
- Most cases occur in teenagers (ages 10-25 years), and patients with **familial retinoblastoma** have a high risk
- Secondary osteosarcomas occur in elderly persons.
 - These highly aggressive tumors are associated with Paget disease, irradiation, and chronic osteomyelitis.



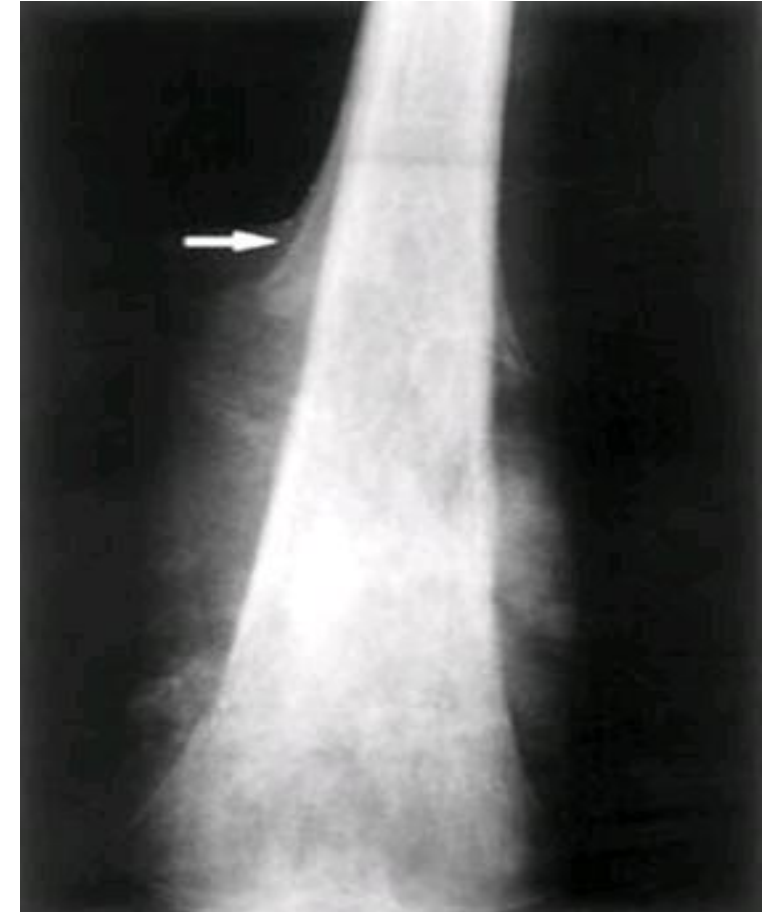
Osteosarcoma

Clinically

- localized pain and swelling.
- The classic x-ray findings are
 - Codman triangle (periosteal elevation),
 - 'sunburst pattern, and bone destruction
- The treatment is with surgery and chemotherapy.
- The prognosis is poor.

Pathology

- It often involves the **metaphyses** of long bones, usually around the knee (**distal femur and proximal tibia**).
- The tumor produces a large, firm, white-tan mass with necrosis and hemorrhage.



Chondrosarcoma

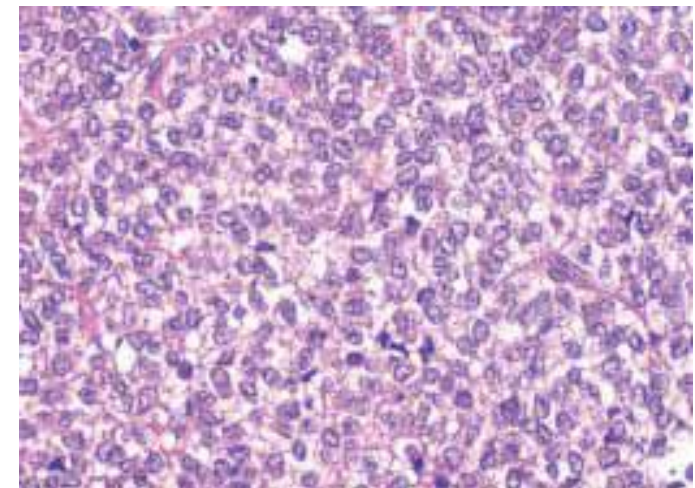
- is a malignant tumor of chondroblasts
- may arise de novo or secondary to a pre-existing enchondroma or Paget disease.
- Males are affected more frequently than females with a peak of 30-60 years.
- presents with enlarging mass with pain and swelling, and it typically involves the
 - pelvic bones, spine, and shoulder girdle.

Ewing sarcoma

- Is a malignant neoplasm of undifferentiated cells arising within the marrow cavity.
- M>F
- Most cases occur in teenagers (age range 15-20)

Clinically

- Patients present with pain, swelling, and tenderness.
- X-ray studies show concentric "**onion-skin**" layering of new periosteal bone.
- The tumor is treated with chemotherapy, surgery, and/or radiation

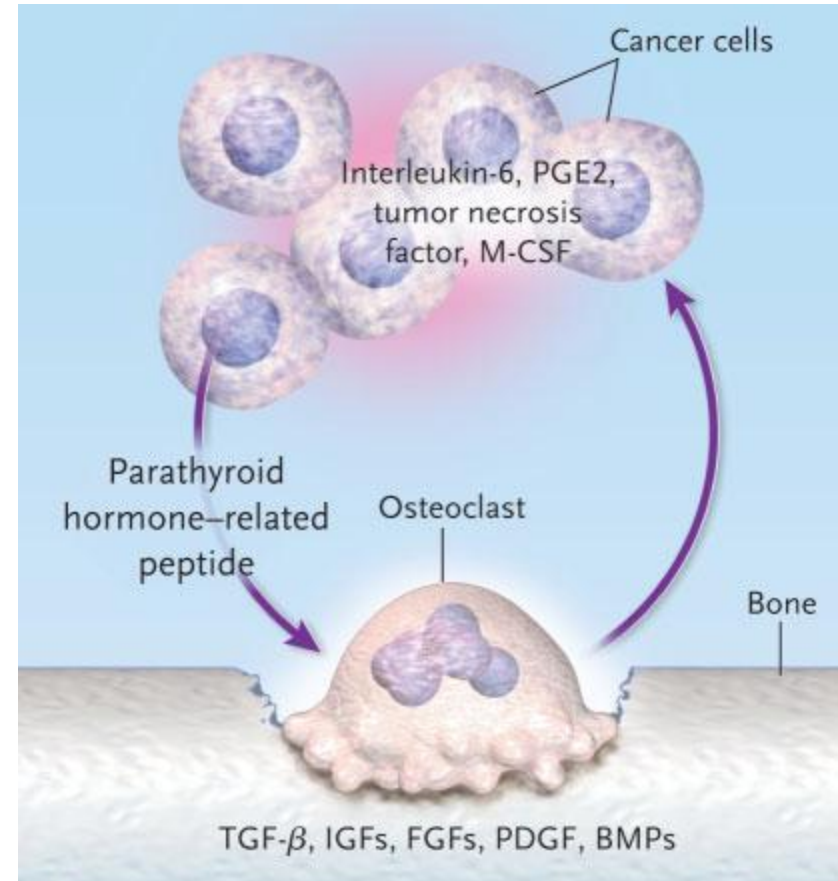


Metastatic Tumors

- The most Common Malignant Tumors in Bone
- Most metastatic lesions to bone are carcinomas, particularly of the breast, prostate, lung, thyroid and kidney.
- hematogenous skeletal metastases are found in at least 85% of cancer
- The **vertebral column** is, by far, the most common site in **adults**, and the **appendicular skeleton** is the typical location in **children**.

Metastatic Tumors

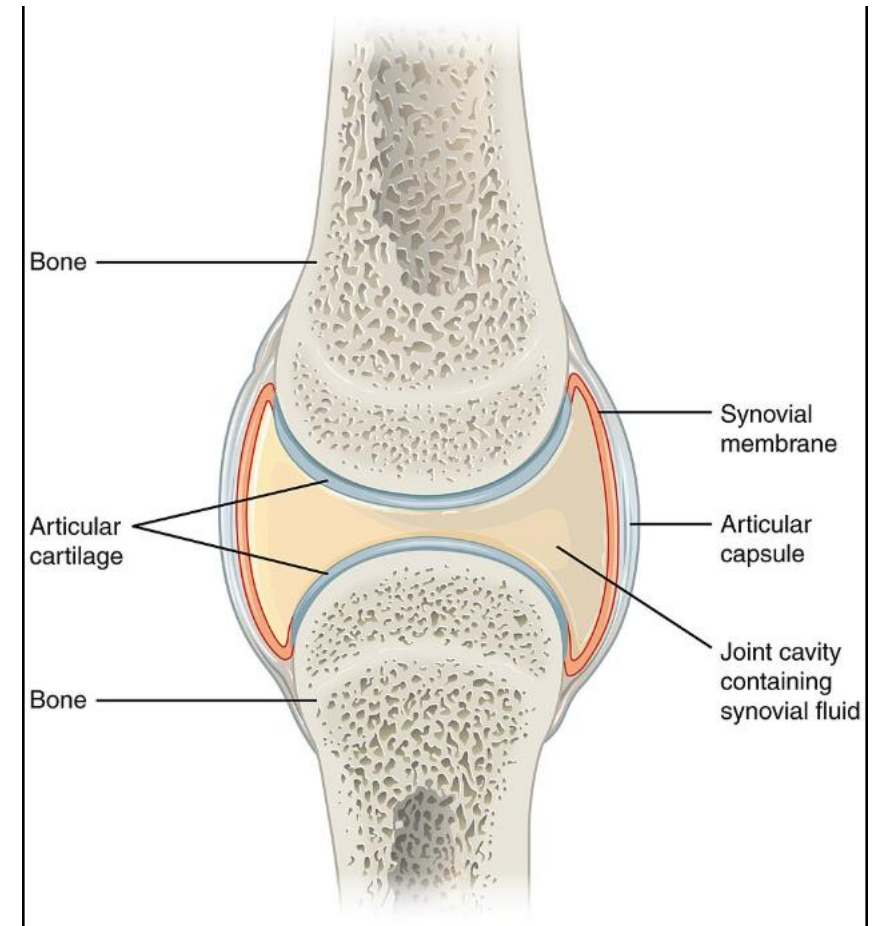
- X-ray
 - Lytic, blastic, or mixed
- Lytic lesions
 - Osteoclast activation
 - prostaglandins, interleukins, and PTHRp
- Blastic lesions
 - stimulate osteoblastic bone formation
 - Prostatic adenocarcinoma



JOINT PATHOLOGY

INTRODUCTION

- A joint (articulation) is a complex structure designed for **movement** and **load-bearing**.
- Key components include:
 - **Articular Cartilage:** Smooth, avascular hyaline cartilage covering bone ends.
 - **Synovium:** Lining membrane that secretes synovial fluid (lubrication/nutrition).
 - **Joint Capsule:** Fibrous tissue providing stability.
 - **Subchondral Bone:** Bone underlying the cartilage.
 - **Synovial Fluid:** Ultrafiltrate of plasma with added hyaluronic acid.
- Pathology typically arises from
 - **degeneration, inflammation,**
 - **metabolic deposition, or infection.**

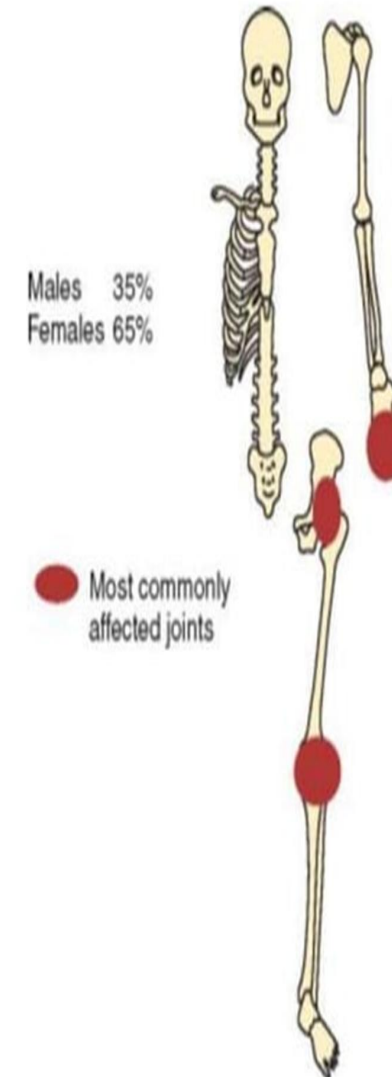


JOINT DISEASES

- Joints are subject to a wide variety of disorders
 - ❖ Degeneration
 - ❖ Infections
 - ❖ immune-mediated injury
 - ❖ metabolic derangements
 - ❖ Neoplasms
- ARTHRITIS
 - ❖ Degenerative (OSTEOARTHRITIS)
 - ❖ Rheumatoid
 - ❖ Infectious: Supp., TB, Lyme, Viral
 - ❖ Gout (URATE)
 - ❖ Pseudogout (PYROPHOSPHATE)
- TUMORS
 - ❖ Ganglion (Synovial Cyst)
 - ❖ Giant Cell Tumor (Pigmented Villo Nodular Synovitis[PVNS]), Synovial Sarcoma

OSTEOARTHRITIS(Degenerative joint disease)

- Characterized by
 - Degeneration of cartilage that results in structural and functional failure of synovial joints
- An intrinsic disease of cartilage in which chondrocytes respond to biochemical & mechanical stresses resulting in breakdown of the matrix
- Most common joint disease worldwide (affects ~10% of men, 18% of women >60)
- **Primary osteoarthritis**
 - Onset - Insidiously with age and without apparent initiating cause
 - Most commonly affected - joints of the hands, knees, hips, and spine
- **Secondary osteoarthritis**
 - strikes in youth
 - predisposing conditions
 - previous trauma
 - developmental deformity
 - systemic disease : diabetes, hemochromatosis
 - marked obesity

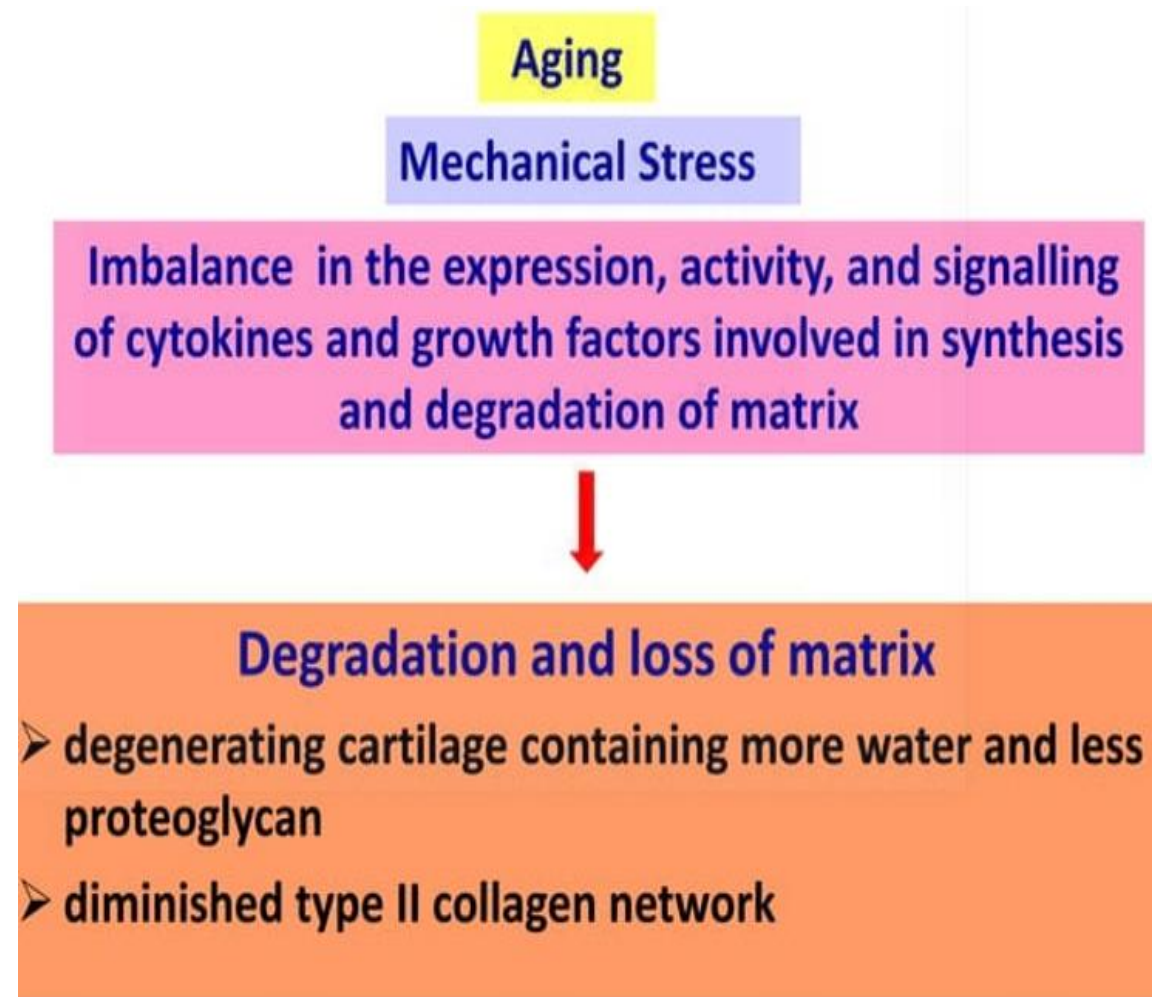


1. Hips
2. Knees
3. lower lumbar
4. cervical vertebrae
5. proximal and distal interphalangeal joints of the fingers
6. first carpometacarpal joints
7. first tarsometatarsal joints of the feet are commonly involved

PATHOGENESIS

Risk Factors

- A wear-and-tear phenomenon
 - Mechanical stresses
 - Aging
 - Genetic factors: polymorphisms and mutations in genes encoding components of the matrix and signaling molecules
 - sustained high estrogen levels
 - increased bone density

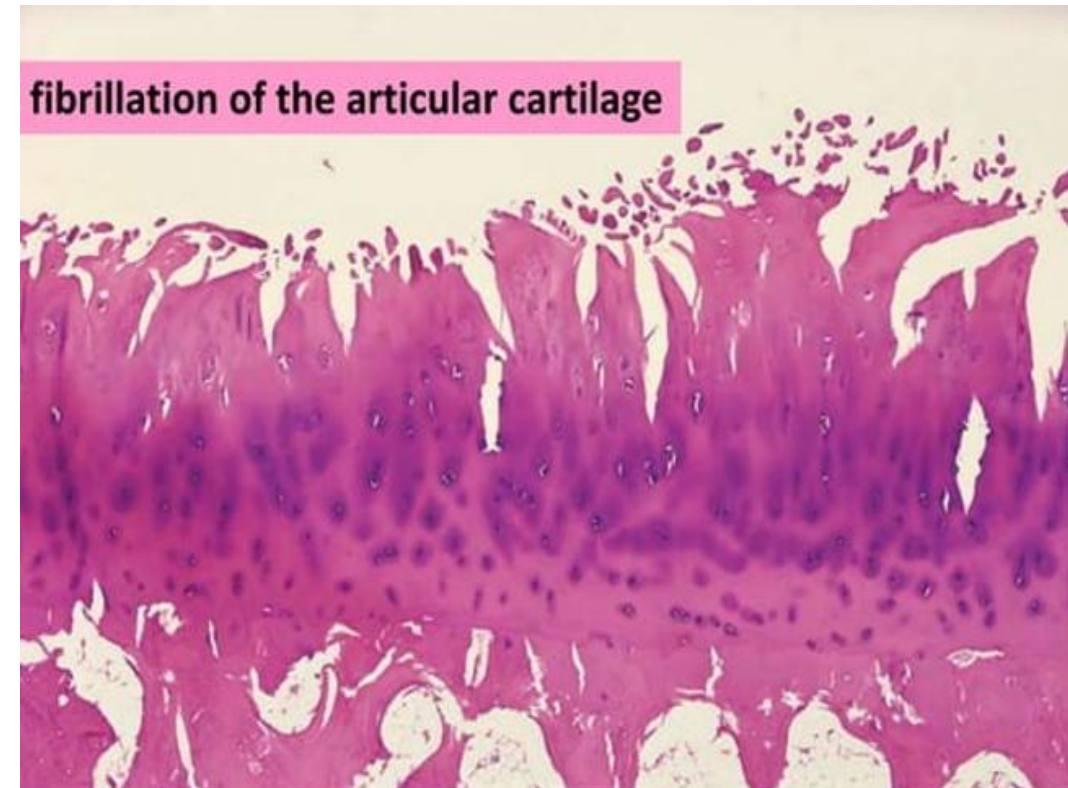


MORPHOLOGY

- Early changes include alterations in the composition and structure of the matrix

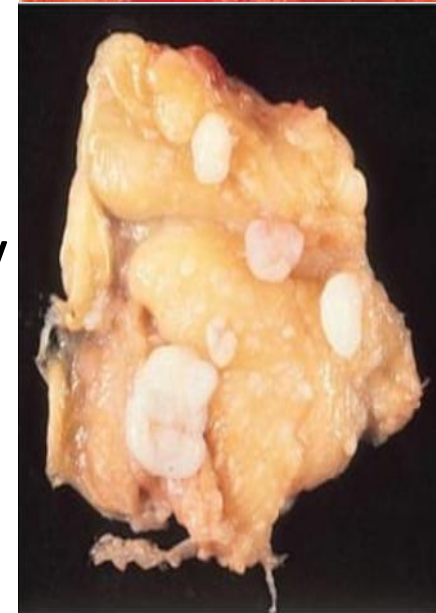
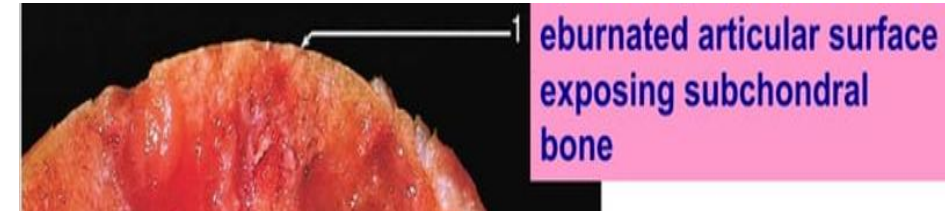
Microscopic:

- vertical and horizontal fibrillation and cracking of the matrix
 - occur as the superficial layers of the cartilage are degraded



Gross morphology

- **Chondromalacia**: at early stage -a soft granular-appearing articular cartilage surface
- **bone eburnation**: Eventually full-thickness portions of the cartilage are lost and the subchondral bone plate is exposed and is smoothed and burnished by friction, giving it the appearance of polished ivory
 - underlying cancellous bone becomes sclerotic and thickened
- **subchondral cyst**: fracture gaps allow synovial fluid to be forced into the subchondral regions to form fibrous walled cysts
- **joint mice**: Small fractures can dislodge pieces of cartilage and subchondral bone into the joint, forming **loose bodies**



cyst in femoral head

- **Osteophytes:** Mushroom-shaped bony outgrowths develop at the margins of the articular surface
- **pannus:** a fibrous synovial pannus covers the peripheral portions of the articular surface in severe disease
- The synovium is usually only mildly congested and fibrotic, and may have scattered chronic inflammatory cells



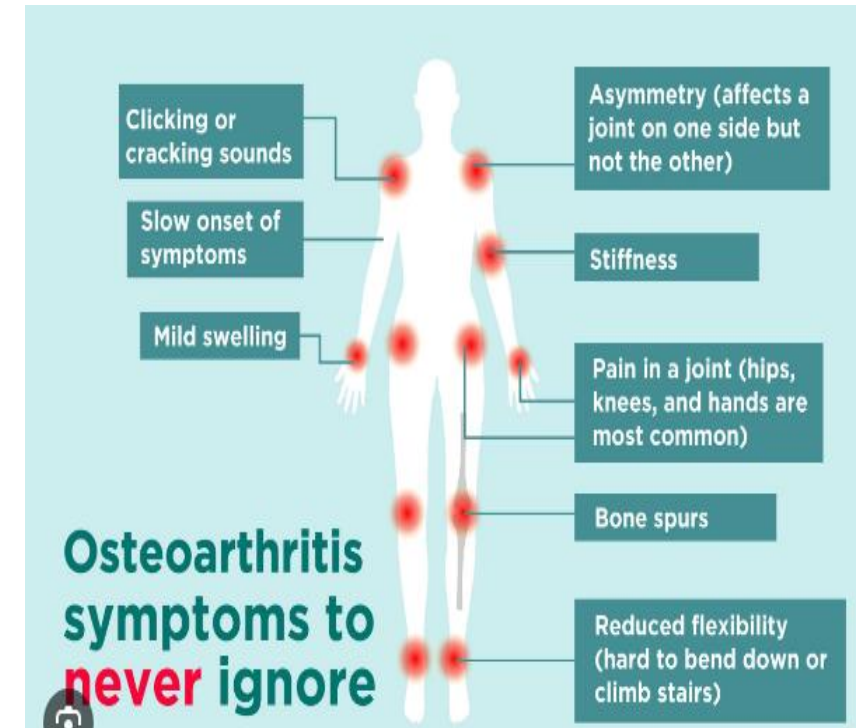
OSTEOARTHRITIS

Clinical Course

- insidious disease
- predominantly affects patients beginning in their **50s and 60s**

Characteristic symptoms and signs

1. deep, aching pain exacerbated by use
2. Morning stiffness
3. crepitus (grating or popping sensation in the joint)
4. Limitation in the range of movement
5. Osteophyte impingement on spinal foramina can cause nerve root compression with radicular pain, muscle spasms, muscle atrophy, and neurologic deficits
6. **Heberden nodes** in the fingers, representing prominent osteophytes at the distal interphalangeal joints are characteristic in women



RHEUMATOID ARTHRITIS

- Systemic **chronic inflammatory** disorder
- Affects many tissues and organs-skin, blood vessels, heart, lungs, and muscles but **principally attacks the joints**
- Causes a **non suppurative proliferative synovitis**
- Progresses to destruction of the articular cartilage and underlying bone resulting in **ankylosis** of the joint

Pathogenesis

- RA is an **autoimmune** disease involving complex interactions of
 - genetic risk factors
 - environment
 - immune system
- Genetic and environmental factors contribute to the **breakdown of tolerance to self-antigens**

RHEUMATOID ARTHRITIS

Pathogenesis

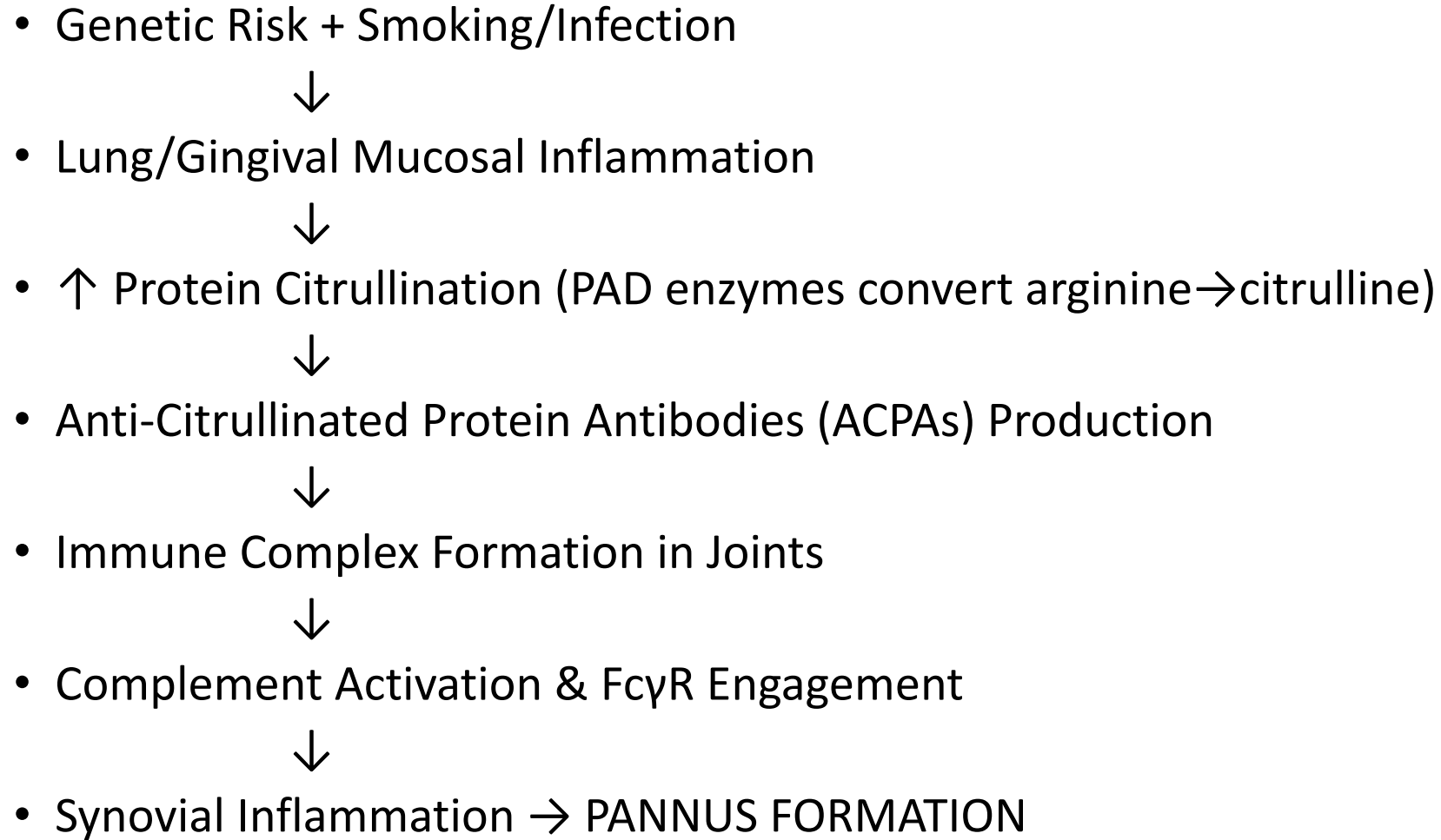
Genetic susceptibility

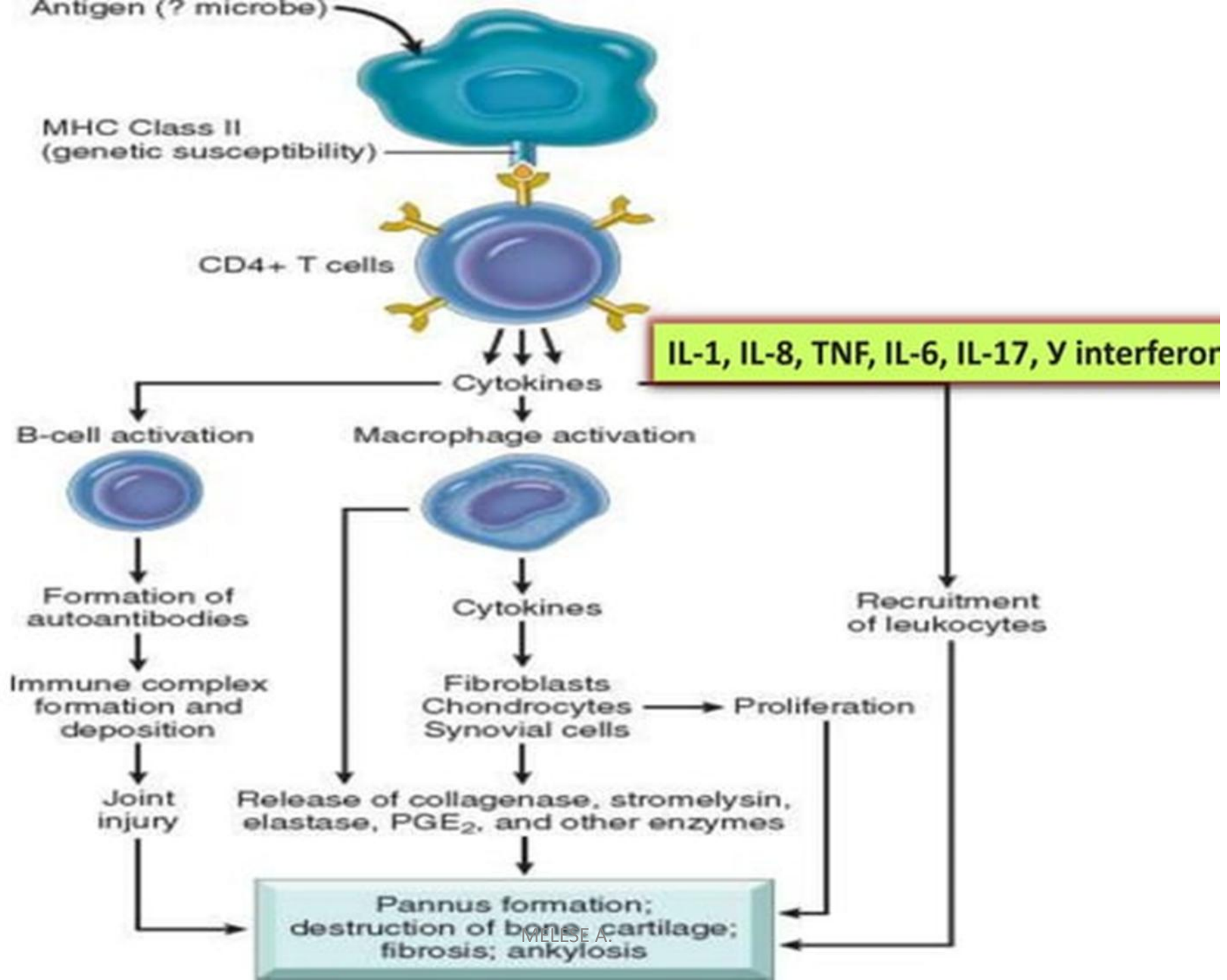
- 50% of the risk of developing RA is related to **genetic factors**
 - Specific HLA-DRB1 alleles
 - PTPN22 gene

ENVIRONMENTAL TRIGGERS

- **Smoking:** Strongest environmental risk (interacts with HLA-DRB1*SE)
- **Microbiome:** Porphyromonas gingivalis (produces PAD enzyme → citrullination)
- **Infections:** EBV, CMV (molecular mimicry)
- **Hormonal:** Postpartum flare, nulliparity

RHEUMATOID ARTHRITIS





RHEUMATOID ARTHRITIS

Morphology

- RA typically manifests as **symmetric** arthritis principally affecting **small joints** of the hands and feet, ankles, knees, wrists, elbows & shoulders
- Most often - **Proximal interphalangeal & metacarpophalangeal joints**
- **Distal interphalangeal joints are spared**
- Axial involvement, when it occurs is limited to the **upper cervical** spine

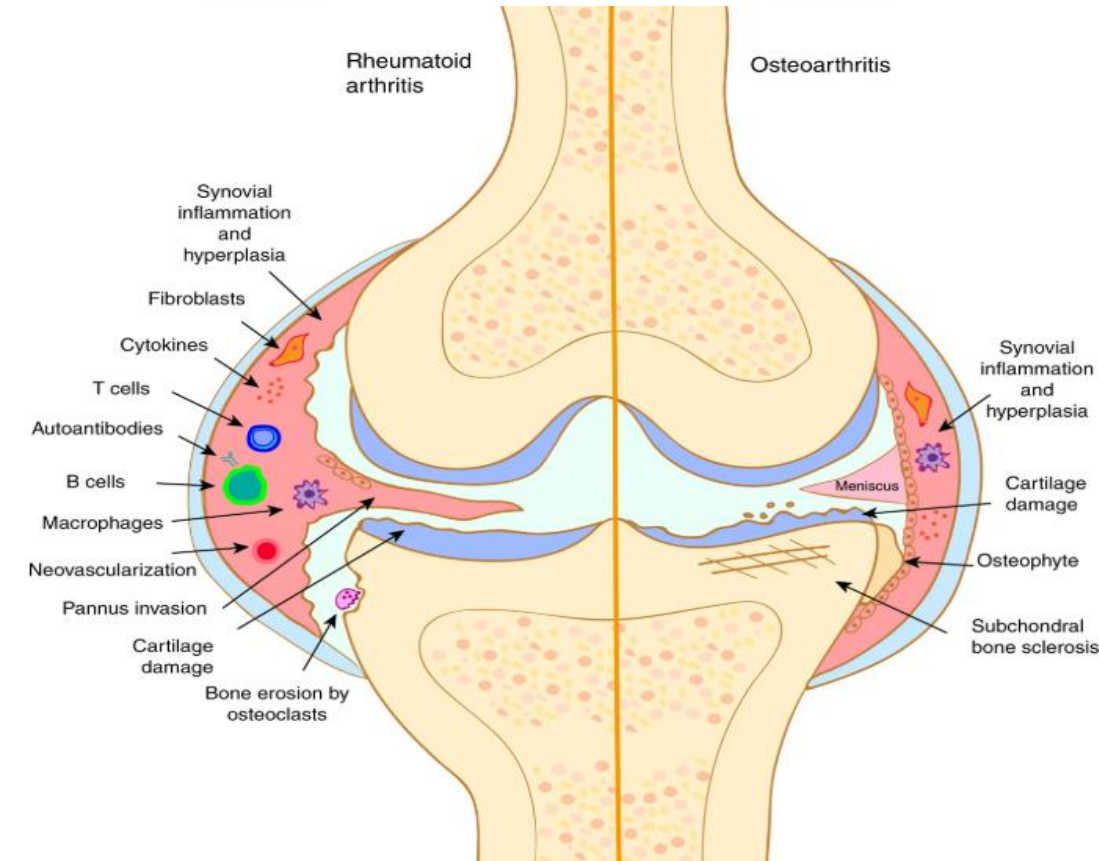
Gross

- synovium becomes
 - Thickened transforming thin, smooth synovial membrane into lush, edematous, frond-like (villous) projections

RHEUMATOID ARTHRITIS

Morphology

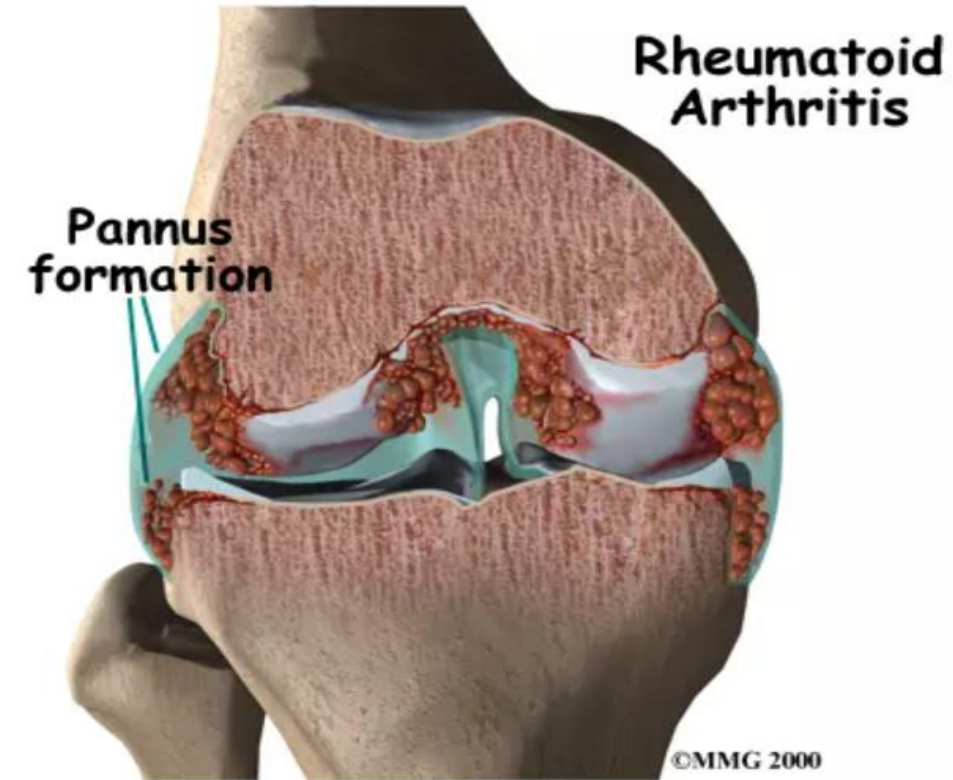
- Affected joints show **chronic papillary synovitis**
- The characteristic histologic features include
 - **Lining Layer Hyperplasia:** >5 cell layers (normal 1-3)
 - **Inflammatory Infiltrate:**
 - **Perivascular lymphocytes** (mainly CD4+ T-cells)
 - **Plasma cells** (often IgG/IgM producers)
 - **Macrophages** (CD68+), some multinucleated
 - **Fibrin Deposition:** On surface and in stroma
 - **Vascular Changes:**
 - **Angiogenesis** (high VEGF expression)
 - **Vasculitis** (mild, not systemic)
 - **Stromal Changes:** Fibrosis, hemosiderin deposition
 - Accumulation of neutrophils
 - osteoclastic activity in underlying bone



RHEUMATOID ARTHRITIS

7. Pannus formation

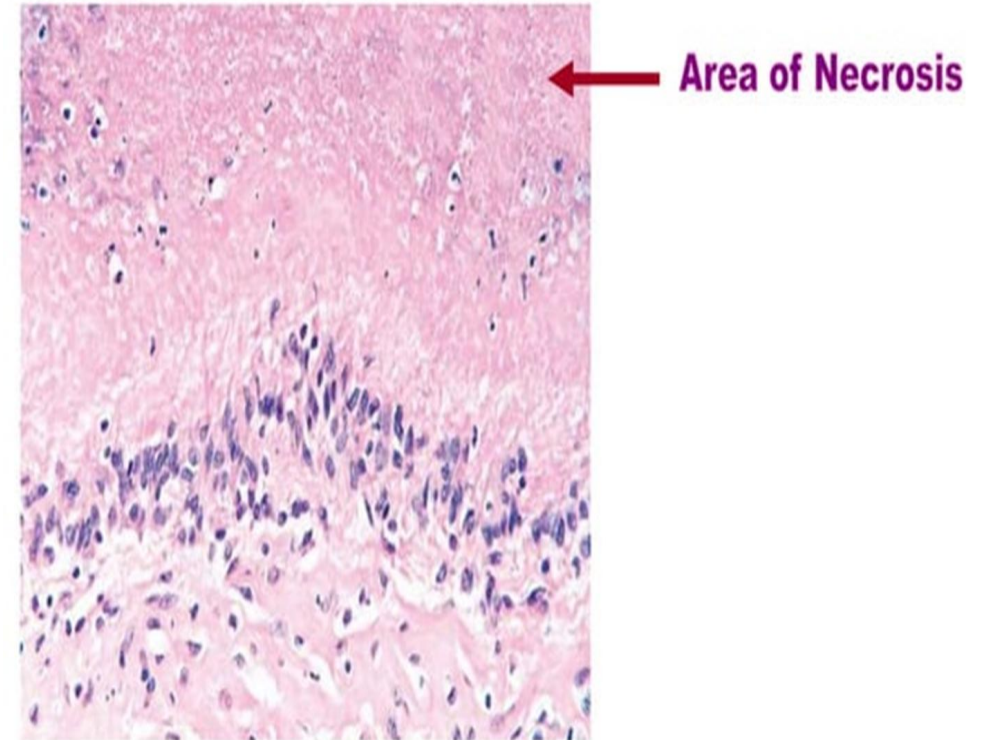
- **THE PANNUS: The Destructive "Tumor"**
- **Definition:** Inflammatory granulation tissue spreading over articular surface
- **Composition:**
 - **Hyperplastic synovial lining** (Type A macrophages, Type B fibroblasts)
 - **Inflammatory cells** (T-cells, plasma cells, macrophages)
 - **New blood vessels** (angiogenesis driven by VEGF)
 - **Osteoclasts** at bone interface
- **Functions:**
 - **Erosion:** Proteases (MMPs, cathepsins) degrade cartilage
 - **Bone Resorption:** RANKL → osteoclast differentiation → juxta-articular erosions
 - **Invasion:** FLS exhibit transformed-like behavior (migration, invasion, apoptosis resistance)



Skin

RHEUMATOID NODULES (20-30%)

- **Pathognomonic** for RA (seropositive patients)
- **Location:** Subcutaneous (extensor surfaces, olecranon), lungs, heart, sclera
- **Histology (3 Zones):**
 - **Central Zone:** Fibrinoid necrosis (pink, amorphous on H&E)
 - **Intermediate Zone:** Palisading macrophages/epithelioid cells (radially arranged)
 - **Outer Zone:** Granulation tissue, lymphocytes, fibrosis
- **Pathogenesis:** Vasculitis → immune complex deposition → necrosis



RHEUMATOID ARTHRITIS

VASCULITIS (10-15% of severe RA)

- **Types:** Leukocytoclastic (small vessels), necrotizing (medium vessels)
- **Manifestations:** Digital infarcts, mononeuritis multiplex, visceral involvement

CARDIOPULMONARY

- **Pleural disease:** Effusions (low glucose, high LDH), pleuritis
- **Lung nodules:** Often cavitory (Caplan syndrome with pneumoconiosis)
- **Interstitial lung disease:** Usual interstitial pneumonia (UIP) pattern
- **Heart:** Pericarditis (most common), valvulitis, coronary arteritis

OCULAR

- **Kerato-conjunctivitis sicca** (Secondary Sjögren's: 30%)
- **Episcleritis/Scleritis** (can progress to scleromalacia perforans)

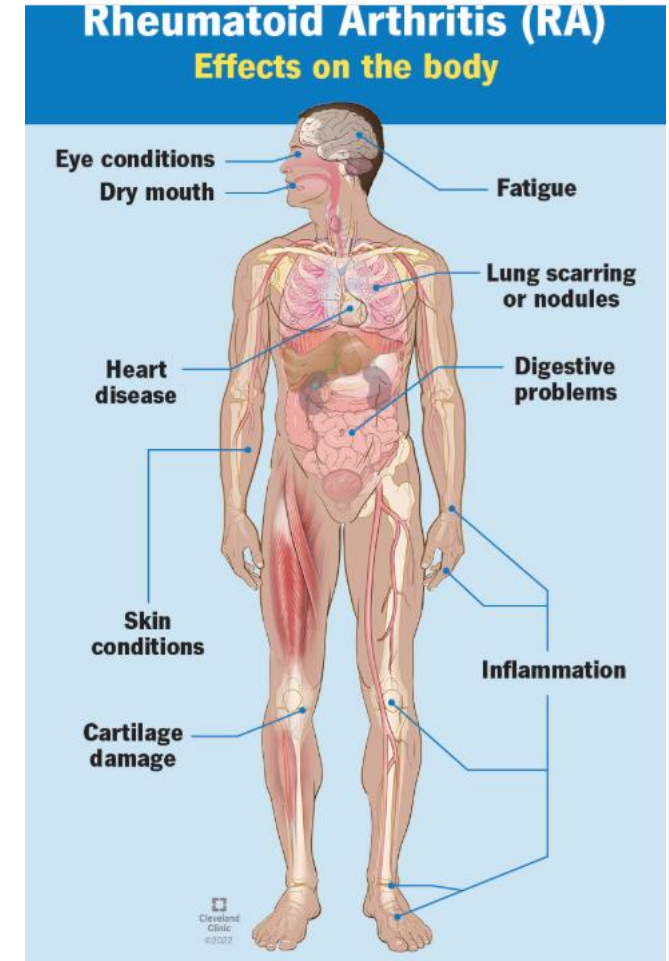
RHEUMATOID ARTHRITIS

Clinical Course

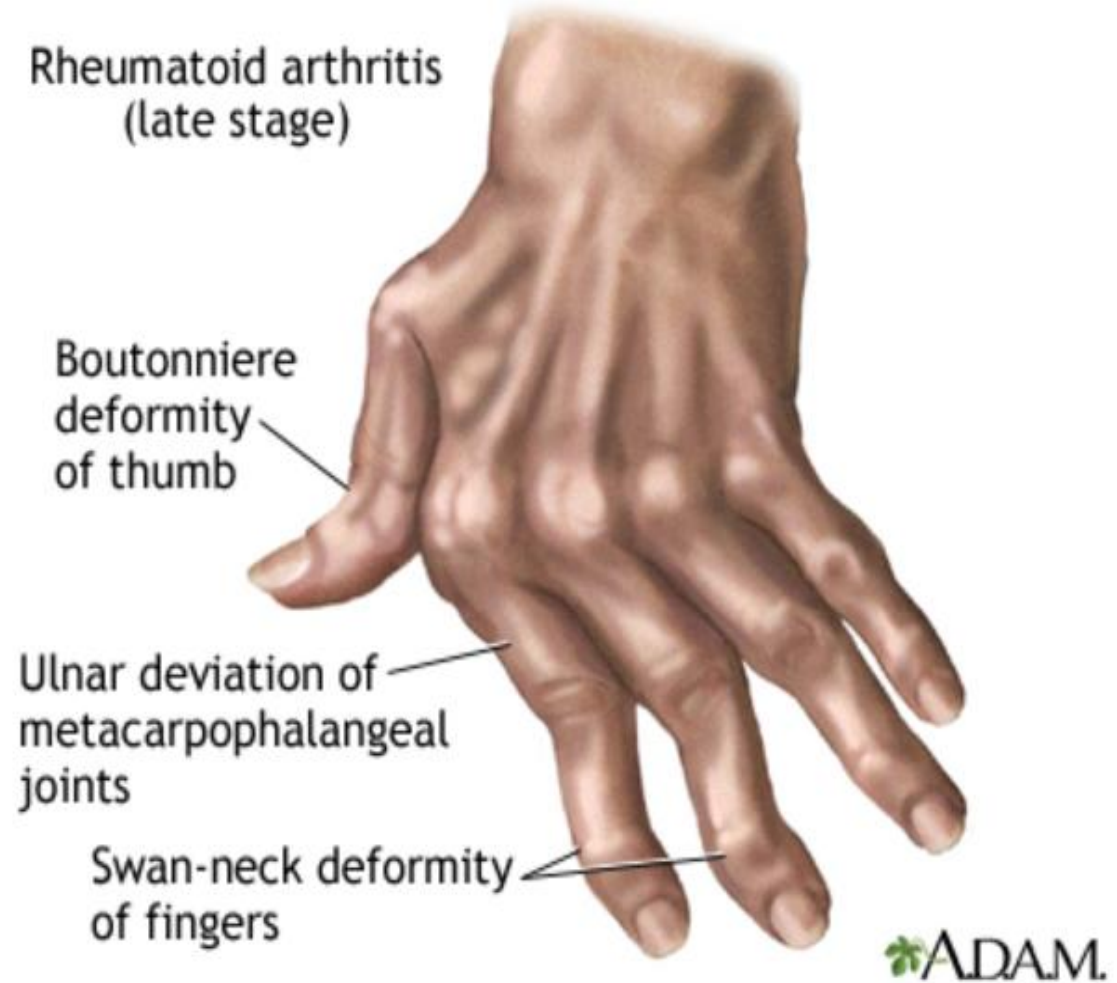
- Symmetric polyarticular arthritis
- Constitutional symptoms (weakness, malaise, and low-grade fever)
- Subcutaneous rheumatoid nodules
- Vasculitic involvement of the extremities may give rise to Raynaud phenomenon and chronic leg ulcers

Symmetric polyarticular arthritis

- The arthritis first appears insidiously with aching and stiffness of the joints, particularly in the morning
- As the disease advances the joints become
 - swollen, painful
 - Limited movement (minimal or no range of motion)
 - complete ankylosis



- characteristic deformities
 - radial deviation of the wrist
 - ulnar deviation of the fingers
 - flexion-hyperextension abnormalities of the fingers (swanneck, boutonnière)
 - Large synovial cysts



RHEUMATOID ARTHRITIS

- The radiographic hallmarks are
 - joint effusions
 - juxta-articular osteopenia
 - erosions and narrowing of the joint space
 - loss of articular cartilage
- The disease course may be slow or rapid, and fluctuates over the years, with the greatest damage occurring in the first 4 or 5 years
- Approximately 20% of affected individuals enjoy periods of partial or complete remission



- Diffuse osteopenia
- Narrow joint spaces
- Periarticular bony erosions
- Ulnar drift of the fingers

Juvenile Rheumatoid Arthritis

- A group of multifactorial disorders with environmental and genetic components
- unknown etiology
- classified according to their presentation into oligoarthritis, polyarthritis & systemic (Still's disease) variants
- Large joints often are affected
- begin before the age of 16 years and persist for more than 6 weeks
- Extraarticular inflammatory manifestations such as uveitis also may be present

GOUT(Crystal-Induced Arthritis)

- Affects about **1%** of the population & shows a predilection for **males**
- Caused by excessive amounts of **uric acid** with in tissues and body fluids
- **Monosodium urate crystals** precipitate from supersaturated body fluids and induce an **acute** inflammatory reaction

Definition

- Gout is marked by **transient** attacks of acute arthritis initiated by **crystallization of monosodium urate** within and around joints.
- Sometimes accompanied by the formation of large **crystalline aggregates** called **tophi**, and eventual permanent joint deformity

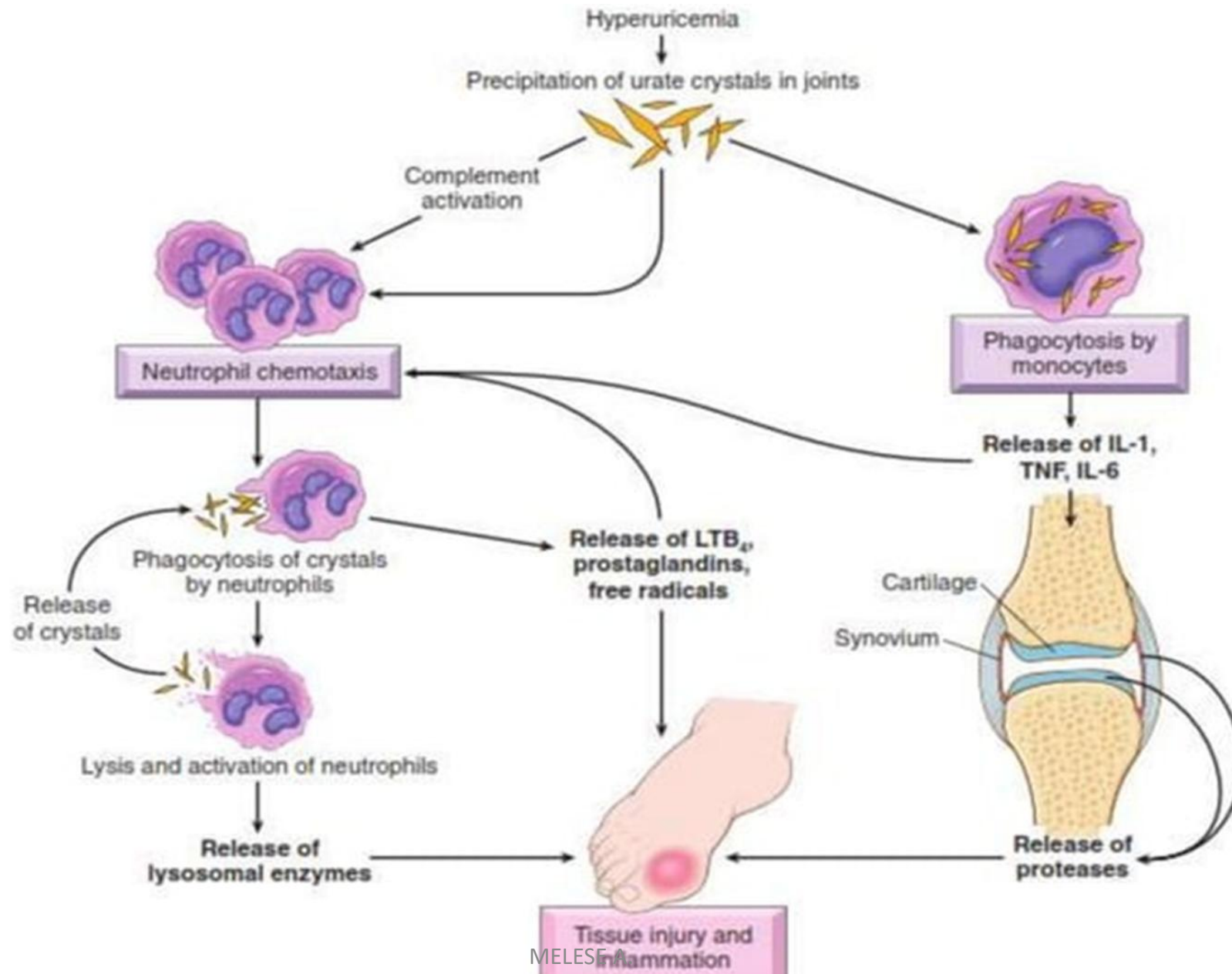
- Gout is divided into primary and secondary forms
- **Primary gout**
 - 90% of cases
 - the basic cause is **unknown** or
 - less commonly, the disorder is due to an **inborn metabolic defect** that causes hyperuricemia
- **Secondary gout**
 - 10% of cases
 - cause of the hyperuricemia is known, but gout is not necessarily the main or even dominant clinical disorder

Clinical Category	Metabolic Defect
Primary Gout (90% of cases)	
Enzyme defects—unknown (85% to 90% of cases)	Overproduction of uric acid Normal excretion (majority) Increased excretion (minority) Underexcretion of uric acid with normal production
Known enzyme defects—e.g., partial HGPRT deficiency (rare)	Overproduction of uric acid
Secondary Gout (10% of cases)	
Associated with increased nucleic acid turnover—e.g., leukemias	Overproduction of uric acid with increased urinary excretion
Chronic renal disease	Reduced excretion of uric acid with normal production
Inborn errors of metabolism	Overproduction of uric acid with increased urinary excretion, e.g., complete HGPRT deficiency (Lesch-Nyhan syndrome)

HGPRT, hypoxanthine guanine phosphoribosyl transferase.

Pathogenesis

- **Urate Handling:**
 - **Production:** 2/3 endogenous, 1/3 dietary
 - **Excretion:** 70% renal, 30% intestinal
- **Causes of Hyperuricemia:**
 - **Underexcretion (90%):** Renal impairment, drugs, genetics
 - **Overproduction (10%):** Enzyme defects, myeloproliferative disorders
- **Critical Concept:** Solubility limit = **6.8 mg/dL** → supersaturation → crystal nucleation



Gout

MORPHOLOGY

- Major morphologic manifestations of gout are
 - acute arthritis
 - chronic tophaceous arthritis
 - tophi in various sites
 - gouty nephropathy

Acute arthritis :

- characterized by **dense neutrophilic infiltrate** permeating the synovium and synovial fluid
- presence of long, slender, **needle-shaped** monosodium urate crystals in the cytoplasm of the neutrophils & in small clusters in the synovium
- synovium is edematous and congested & contains scattered mononuclear inflammatory cells
- when the episode of crystallization abates and the crystals resolubilize, the attack remits

Gout

Chronic tophaceous arthritis evolves from

- Repetitive precipitation of urate crystals during acute attacks
- urates can heavily encrust the articular surfaces and form visible deposits in the synovium
- In severe cases, fibrous or bony ankylosis ensues, resulting in loss of joint function

Tophi in various sites

- Tophi are pathognomonic for gout
 - formed by large aggregations of urate crystals surrounded by an intense inflammatory reaction of lymphocytes, macrophages, & foreign-body giant cells, attempting to engulf the masses of crystals
- Sites
 - articular cartilage of joints
 - periarticular ligaments, tendons, & soft tissues
 - ear lobes, nasal cartilages, skin of the finger tips
- Superficial tophi can lead to large ulcerations of the overlying skin

Gout

RENAL MANIFESTATIONS

- **Urate Nephropathy:** Intratubular crystal deposition (acidic, concentrated urine)
 - **Pathology:** Needle-shaped crystals in collecting ducts → tubular obstruction + interstitial inflammation
- **Uric Acid Stones:** 10-25% of gout patients (**radiolucent**)
- **Chronic Urate Nephropathy:** Arteriolar nephrosclerosis (hypertension association)

Gout

Clinical Features

- More common in **men** than in women
- Does not usually cause symptoms before the age of 30

Risk factors:-

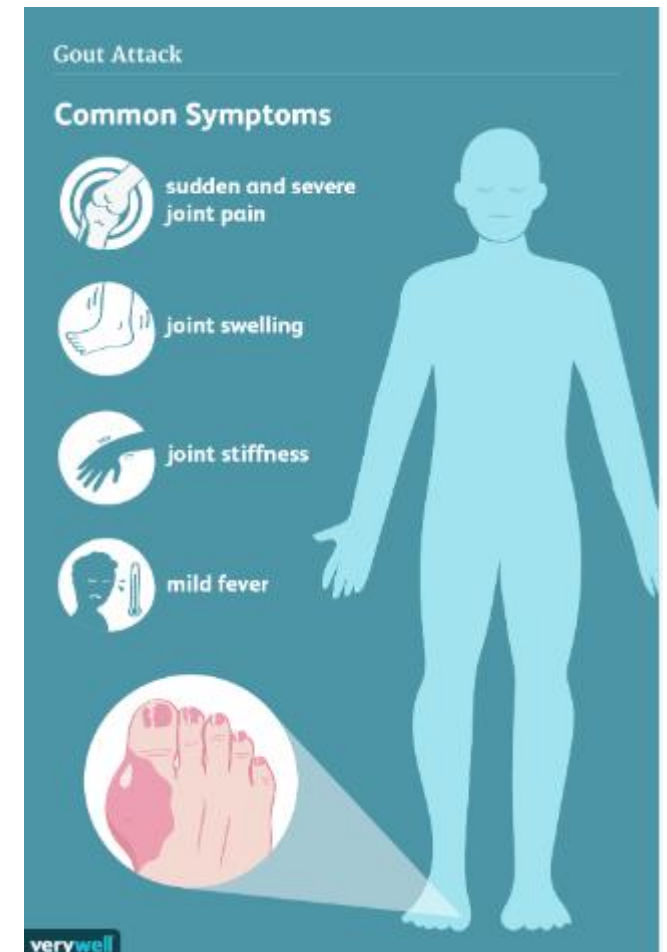
- Obesity
- excess alcohol intake
- consumption of purine-rich foods
- Diabetes
- metabolic syndrome
- renal failure

Gout

- Four stages are classically recognized:
 1. asymptomatic hyperuricemia:
 - around puberty in males and after menopause in women
 2. acute gouty arthritis
 3. "inter-critical" gout (asymptomatic inter-critical period)
 4. chronic tophaceous gout

Acute gouty arthritis:

- Appears after a long interval of years
- sudden onset, excruciating joint pain with localized erythema & warmth
- Mostly monoarticular
- >50% occur in the **first metatarsophalangeal joint(great toe)** & 90% in ankle, heel, or wrist
- last for hours to weeks & gradually resolves, and the patient enters an asymptomatic inter-critical period



Gout

Chronic tophaceous gout

- At this stage, radiographs show characteristic
 - juxtaarticular bone erosion caused by the crystal deposits and loss of the joint space
- Progression leads to chronic gouty nephropathy
 - renal colic associated with the passage of gravel and stones

Pseudogout

- **Calcium pyrophosphate** crystal deposition disease(also known as **chondrocalcinosis**)
- crystal deposits first appear in structures composed of cartilage such as menisci, intervertebral discs, and articular surfaces
- Crystals: Small, **rhomboid/rod-shaped**
- When the deposits enlarge enough, they may rupture, inducing an inflammatory reaction
- Typically first occurs in persons 50 years of age or older becoming more common with increasing age, and eventually reaching a prevalence of 30% to 60% in those age 85
- No gender or race predilection

Infectious Arthritis

- Mode of entry:
 - hematogenous dissemination
 - direct inoculation
 - contiguous spread from osteomyelitis or a soft tissue abscess
- Infectious arthritis is serious because it can cause rapid joint destruction and permanent deformities

Suppurative Arthritis

Etiology (causative agent)

- In children < 2 years of age - Hemophilus influenzae
- In older children and adults - S. aureus & gonococcus
- Patients with sickle cell disease - Salmonella infection at any age
- Both genders are affected equally except for gonococcal arthritis, which occurs mainly in **sexually active women**
- In this group, those with deficiency of complement proteins (C5, C6, C7) are particularly susceptible to disseminated gonococcal infections

Suppurative Arthritis

Clinical presentation

- sudden onset with cardinal signs of acute inflammation- Fever, leukocytosis, and elevated ESR
- involves only a single joint-usually the knee- followed in order by hip, shoulder, elbow, wrist, and sternoclavicular joints
- Joint aspiration typically yields a purulent fluid in which the causal agent can be identified



THANK YOU!